

Generalized linear models

Generalized linear models - Basic concept

Prof. Dr. Andreas Groll

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GLM Basic concept - Contents

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Generalized linear model

Basic concept

What did the previous models have in common?

What is modeled?

expected value $E[y|x]$ or

$\log\left(\frac{\pi}{n-\pi}\right)$, respectively

How is it modeled?

via the linear predictor $x^T \beta$

Basic concept

A bit more formal:

1. The expectation $\mu = \mathbb{E}(y \mid \mathbf{x})$ is linked with the linear predictor $\eta = \mathbf{x}^\top \beta$.
 $\mu = h(\eta)$ $h \hat{=}$ response function, strictly mon. increasing
 $\eta = g(\mu)$ $g = h^{-1} \hat{=}$ link function, 2x differentiable
2. The distribution of y can be represented as a single-parameter exponential family.

NV ✓
Bernoulli - dist. ✓

Exponential family

Single-parameter exponential family - GLM representation

$$f(y_i | \vartheta_i) = \exp \left(\frac{y_i \vartheta_i - b(\vartheta_i)}{\phi} w_i + c(y_i, \phi, w_i) \right)$$

ϑ_i Parameter of interest; natural/canonical parameter

ϕ Dispersion parameter, only of subsidiary interest
(nuisance parameter), independent from observation i

$b(\cdot)$ Twice differentiable function

$c(\cdot)$ Normalization function

w_i known weight, usually $w_i = 1$ (individual data); $w_i > 1$ for grouped data

Exponential family

Example: Normal distribution

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \quad \text{with } \varepsilon_i \sim N(0, \sigma^2), \quad i = 1, \dots, n$$

$$\Leftrightarrow y_i \mid \mathbf{x}_i \sim N(\mathbf{x}_i^\top \boldsymbol{\beta}, \sigma^2)$$

1. Linkage of expectation and $\mathbf{x}^\top \boldsymbol{\beta}$:

$$\mu_i(\mathbf{x}_i) = \mathbb{E}[y_i \mid \mathbf{x}_i] = \mathbf{x}_i^\top \boldsymbol{\beta} = \text{id}(\mathbf{x}_i^\top \boldsymbol{\beta})$$

$$h = \text{id}, \quad g = \text{id}$$

$\Rightarrow$ both response & link function are given by the identity function

Exponential family

Example: Normal distribution

2. Exponential family:

$$\begin{aligned} f(y_i | \mathbf{x}_i) &= \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{(y_i - \overbrace{\mathbf{x}_i^\top \boldsymbol{\beta}}^{\mu_i})^2}{2\sigma^2} \right) \\ &= \exp(-\log(\sqrt{2\pi}\sigma)) \cdot \exp \left(\frac{-y_i^2 + 2\mu_i y_i - \mu_i^2}{2\sigma^2} \right) \\ &= \exp \left(\frac{\mu_i y_i - \frac{\mu_i^2}{2}}{\sigma^2} - \frac{y_i^2}{2\sigma^2} - \log(\sqrt{2\pi}\sigma) \right) \\ \eta_i &= \mu_i \quad (\text{function of } \mu_i; \eta_i(\mu_i) = \mu_i) \\ \phi &= \sigma^2 \quad (\text{dispersion / variation}) \\ b(\eta_i) &= b(\mu_i) = \frac{\mu_i^2}{2} \\ c(y_i, \phi) &= -\frac{y_i^2}{2\sigma^2} - \log(\sqrt{2\pi}\sigma) \end{aligned}$$

Exponential family

$$\mathcal{B}(1, \pi(x_i))$$

Example: Bernoulli distribution (binary response)

1. Linkage of expectation and $\mathbf{x}^T \beta$: $y_i | x_i \sim \text{Ber}(\pi(x_i))$

$$\mu_i(x_i) = \mathbb{E}[y_i | x_i] = \pi(x_i) = \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)}$$

(logistic model)

$$h(\eta) = \frac{\exp(\eta)}{1 + \exp(\eta)} = \frac{\exp(x^T \beta)}{1 + \exp(x^T \beta)} \quad \text{response fct.}$$

$$g(\mu) = g(\pi) = \log\left(\frac{\pi}{1-\pi}\right) \quad \text{link function}$$

Exponential family

Example: Bernoulli distribution (binary response)

2. Exponential family: $f(y_i | \underline{x}_i) = \pi(\underline{x}_i)^{y_i} (1 - \pi(\underline{x}_i))^{1-y_i}$

$$= \exp(y_i \log(\pi(\underline{x}_i)) + (1-y_i) \log(1-\pi(\underline{x}_i)))$$

$$= \exp\left(y_i \log\left(\frac{\pi(\underline{x}_i)}{1-\pi(\underline{x}_i)}\right) + \log(1-\pi(\underline{x}_i))\right)$$

$$v_i = \log\left(\frac{\pi(\underline{x}_i)}{1-\pi(\underline{x}_i)}\right) \Rightarrow v_i = \underline{x}_i^T \beta$$
$$\Rightarrow v_i(\mu_i) = \log\left(\frac{\mu_i}{1-\mu_i}\right)$$

$$\phi = 1, \quad c(y_i | \phi) = 0$$

$$b(v_i) = -\log(1-\pi(\underline{x}_i)) = -\log\left(1 - \frac{\exp(\underline{x}_i^T \beta)}{1 + \exp(\underline{x}_i^T \beta)}\right)$$
$$\Rightarrow \log\left(\frac{1}{1 + \exp(\underline{x}_i^T \beta)}\right) = \log\left(1 + \exp(\underline{x}_i^T \beta)\right)$$

Exponential family

Other distributions

- ▶ Sometimes the normal distribution seems not adequate, since $y_i \geq 0$ (e.g., income)

- ▶ Then more sensible: $\mu_i = \exp(\underline{x}_i^T \beta) > 0$ Resp. Fct.

$$\Leftrightarrow \eta_i = \underline{x}_i^T \beta = \log(\mu_i) \quad \text{link function}$$

- such a model is also called log-linear model
- log-expectation is determined linearly

General remarks

- ▶ For the normal distribution, expectation and variance of the distribution are determined by two parameters; for the Bernoulli distribution, the distribution is determined by just one parameter.
- ▶ A GLM is determined by the choice of
 - ▶ Response / link function
 - ▶ Type of exponential family
 - ▶ Covariates

Generalized linear model

Generalized linear model (GLM)

A GLM is a regression model that fulfills the following assumptions:

- (1) **Structural assumption:** Linkage of expectation $\mu = \mathbb{E}(y \mid \mathbf{x})$ and linear predictor $\mathbf{x}^\top \boldsymbol{\beta}$ via response/ link funktion
- (2) **Distribution assumption:** Density of $y \mid \mathbf{x}$ can be rearranged as single-parameter exponential family.

Generalized linear model

Canonical link

$h(\cdot)$ and $g(\cdot)$ are called **canonical** response or link function, if

$$\vartheta_i = \mathbf{x}_i^\top \boldsymbol{\beta},$$

i.e. the natural parameter ϑ_i is determined linearly.

Examples:

- ▶ Normal distribution: $\vartheta_i = \mu_i = \underline{\mathbf{x}}_i^\top \underline{\boldsymbol{\beta}} = \eta_i$ canonical
 - ▶ Bernoulli distribution: $\vartheta_i = \log\left(\frac{\pi_i}{1-\pi_i}\right) = \underline{\mathbf{x}}_i^\top \underline{\boldsymbol{\beta}}$ canonical
- $\Leftrightarrow \pi_i = \frac{\exp(\underline{\mathbf{x}}_i^\top \underline{\boldsymbol{\beta}})}{1 + \exp(\underline{\mathbf{x}}_i^\top \underline{\boldsymbol{\beta}})} \Rightarrow \text{logistic model}$

Generalized linear model

Why is the assumption of the exponential family important?

→ Statistical inference can be done in a unique framework.

Characteristics of an exponential family

$$(a) \quad \mathbb{E}[y_i | \mathbf{x}_i] = \mu_i = b'(\vartheta_i) = \frac{\partial b(\vartheta_i)}{\partial \vartheta_i}$$

$$(b) \quad \text{Var}(y_i | \mathbf{x}_i) = \phi \cdot b''(\vartheta_i) = \phi \cdot \frac{\partial^2 b(\vartheta_i)}{\partial \vartheta_i^2}$$

Generalized linear model

Proof: for (a)

For $w_i = 1$, the density yields:

$$\begin{aligned}f(y_i | \vartheta_i) &= \exp \left(\frac{y_i \vartheta_i - b(\vartheta_i)}{\phi} + c(y_i, \phi) \right) \\&= \exp \left(\frac{y_i \vartheta_i}{\phi} \right) \exp \left(\frac{-b(\vartheta_i)}{\phi} \right) \exp(c(y_i, \phi))\end{aligned}$$

We have : $\int_{\mathbb{R}} f(y_i | \vartheta_i) dy_i = 1$ (density)

$$\Leftrightarrow \exp \left(\frac{-b(\vartheta_i)}{\phi} \right) \int_{\mathbb{R}} \exp \left(\frac{y_i \vartheta_i}{\phi} \right) \exp(c(y_i, \phi)) dy_i = 1$$

$$\stackrel{\log(\dots)}{\Leftrightarrow} -\frac{b(\vartheta_i)}{\phi} + \log \left(\int_{\mathbb{R}} \exp \left(\frac{y_i \vartheta_i}{\phi} \right) \exp(c(y_i, \phi)) dy_i \right) = 0$$

Generalized linear model

Derivative w.r.t. η_i :

$$- \frac{b'(\eta_i)}{\phi} + \frac{\frac{\partial}{\partial \eta_i} \left(\int_{\mathbb{R}} \exp\left(\frac{y_i \eta_i}{\phi} + c(y_i, \phi)\right) dy_i \right)}{\int_{\mathbb{R}} \exp\left(\frac{y_i \eta_i}{\phi} + c(y_i, \phi)\right) dy_i} = 0$$

$$\begin{aligned} (*)_{\text{rule}} &= \int_{\mathbb{R}} \left[\frac{\partial}{\partial \eta_i} \exp\left(\frac{y_i \eta_i}{\phi} + c(y_i, \phi)\right) \right] dy_i \\ &= \int_{\mathbb{R}} \left[\exp\left(\frac{y_i \eta_i}{\phi} + c(y_i, \phi)\right) \frac{y_i}{\phi} \right] dy_i \end{aligned}$$

$$\begin{aligned} (*) \Rightarrow b'(\eta_i) &= \frac{\int_{\mathbb{R}} \left[\exp\left(\frac{y_i \eta_i}{\phi} + c(y_i, \phi)\right) y_i \right] dy_i}{\int_{\mathbb{R}} \exp\left(\frac{y_i \eta_i}{\phi} + c(y_i, \phi)\right) dy_i} \cdot \frac{\phi}{\exp\left(\frac{y_i \eta_i}{\phi}\right)} \\ &= \frac{\int_{\mathbb{R}} f(y_i | \eta_i) y_i dy_i}{\int_{\mathbb{R}} f(y_i | \eta_i) dy_i} = \frac{\mathbb{E}[y_i | x_i]}{1} = \mu_i \quad \square \end{aligned}$$

Generalized linear model

Example: Normal distribution

$$y_i = \mu_i, \quad b(\mu_i) = b(\mu_i) = \frac{\mu_i^2}{2}, \quad \phi = \sigma^2$$

$$\Rightarrow b'(\mu_i) = \frac{2\mu_i}{2} = \mu_i = E[y_i | x_i]$$

$$\Rightarrow b''(\mu_i) = 1 = v(\mu_i)$$

$$\Rightarrow \text{var}(y_i | x_i) = \phi \cdot b''(\mu_i) = \sigma^2 \cdot 1 = \sigma^2$$

Generalized linear model

Example: Bernoulli distribution

$$b(\eta_i) = \log(1 + \exp(\eta_i)), \quad \phi = 1$$

$$\Rightarrow b'(\eta_i) = \frac{1}{1 + \exp(\eta_i)} \cdot \exp(\eta_i) = \pi_i = E[y_i | x_i]$$

$$\Rightarrow b''(\eta_i) = \frac{e^{\eta_i}(1 + e^{\eta_i}) - e^{\eta_i} \cdot e^{\eta_i}}{(1 + e^{\eta_i})^2} = \frac{e^{\eta_i}}{(1 + e^{\eta_i})^2}$$

$$= \frac{e^{\eta_i}}{1 + e^{\eta_i}} \cdot \frac{1}{1 + e^{\eta_i}} = \pi_i (1 - \pi_i)$$

$$\Rightarrow \text{var}(y_i | x_i) = \phi \cdot b''(\eta_i) = \pi_i (1 - \pi_i)$$